SENTIMENT ANALYSIS OF X (Twitter) DATA

1.INTRODUCTION:

Social media platforms such as X (formerly Twitter) have become the primary space for public conversations. Millions of posts are shared daily, expressing views on politics, society, technology, and global events. Analyzing this data manually is not possible due to its scale and diversity. Sentiment Analysis, a Natural Language Processing (NLP) technique, allows us to automatically classify posts as positive, negative, or neutral.

The objective of this internship project was to apply sentiment analysis on a dataset of X posts to identify prevailing public opinions. The task was provided by ShadowFox as part of an internship assignment

2. DATASET:

The dataset was shared in CSV format and uploaded to Google Colab for processing. The dataset contained a column named clean_text representing the preprocessed posts and a category column with sentiment labels (-1 = Negative, 0 = Neutral, 1 = Positive). Since the dataset was large (over 21 MB), it provided sufficient variety for meaningful analysis.

3. DATA CLEANING:

Before applying sentiment analysis, the raw text needed to be cleaned to remove unnecessary characters. The cleaning steps included:

- Removing links (http://...)
- Removing user mentions (@username)
- Removing hashtags (#topic) but keeping the word itself
- Removing numbers, punctuation, and special characters
- Converting text to lowercase for uniformity

This ensured that the input for sentiment analysis was clean and focused on meaningful words.

4. SENTIMENT CLASSIFICATION:

For classification, we used the TextBlob library. Each cleaned post was assigned a polarity score ranging from -1 to +1:

- Polarity > 0 → Positive
- Polarity < 0 → Negative
- Polarity = 0 → Neutral

A new column sentiment was added to the dataset, storing the predicted sentiment for each post.

5. CODE IMPLIMENTATION:

```python
# 1. Install libraries (only needed in Google Colab)
# !pip install pandas numpy matplotlib seaborn nltk textblob wordcloud

# 2. Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import re
from textblob import TextBlob
from wordcloud import WordCloud

# 3. Load dataset
# If running in Google Colab, upload file first:
# from google.colab import files
# uploaded = files.upload()
# df = pd.read_csv("your_dataset.csv")

import pandas as pd
import io
from google.colab import files

# Upload the file
uploaded = files.upload()

# Check the actual filename uploaded
print("Uploaded files:", uploaded.keys())
```

```python
# Use the correct key name here
df = pd.read_csv(io.BytesIO(uploaded["X data (5).csv"]))

print("✅ Dataset Loaded Successfully!")
print("Columns in dataset:", df.columns)
print(df.head())


# 4. Clean tweets
def clean_tweet(text):
    text = re.sub(r"http\S+", "", str(text))      # remove links
    text = re.sub(r"@\w+", "", text)             # remove mentions
    text = re.sub(r"#", "", text)                # remove hashtag symbol
    text = re.sub(r"[^A-Za-z\s]", "", text)      # remove special chars & numbers
    return text.lower().strip()
# Ensure no NaN values and convert to string
df["cleaned_text"] = df["clean_text"].fillna("").astype(str)

# Sentiment classification
def get_sentiment(text):
    analysis = TextBlob(text)
    if analysis.sentiment.polarity > 0:
        return "Positive"
    elif analysis.sentiment.polarity < 0:
        return "Negative"
    else:
        return "Neutral"

df["sentiment"] = df["cleaned_text"].apply(get_sentiment)
```

```python
print(df["sentiment"].value_counts())


# 6. Visualization: Sentiment distribution
plt.figure(figsize=(6,4))
sns.countplot(x="sentiment", data=df, palette="Set2")
plt.title("Sentiment Distribution")
plt.show()

# 7. Visualization: Word Clouds
positive_words = " ".join(df[df["sentiment"]=="Positive"]["cleaned_text"])
negative_words = " ".join(df[df["sentiment"]=="Negative"]["cleaned_text"])

wordcloud_pos = WordCloud(width=800, height=400, background_color="white").generate(positive_words)
plt.imshow(wordcloud_pos, interpolation="bilinear")
plt.axis("off")
plt.title("Positive Words")
plt.show()

wordcloud_neg = WordCloud(width=800, height=400, background_color="white").generate(negative_words)
plt.imshow(wordcloud_neg, interpolation="bilinear")
plt.axis("off")
plt.title("Negative Words")
plt.show()

# 8. (Optional) Sentiment trend over time if 'date' column exists
if "date" in df.columns:
    df["date"] = pd.to_datetime(df["date"], errors="coerce")
```


```python
wordcloud_neg = WordCloud(width=800, height=400, background_color="white").generate(negative_words)
plt.imshow(wordcloud_neg, interpolation="bilinear")
plt.axis("off")
plt.title("Negative Words")
plt.show()

# 8. (Optional) Sentiment trend over time if 'date' column exists
if "date" in df.columns:
    df["date"] = pd.to_datetime(df["date"], errors="coerce")
```

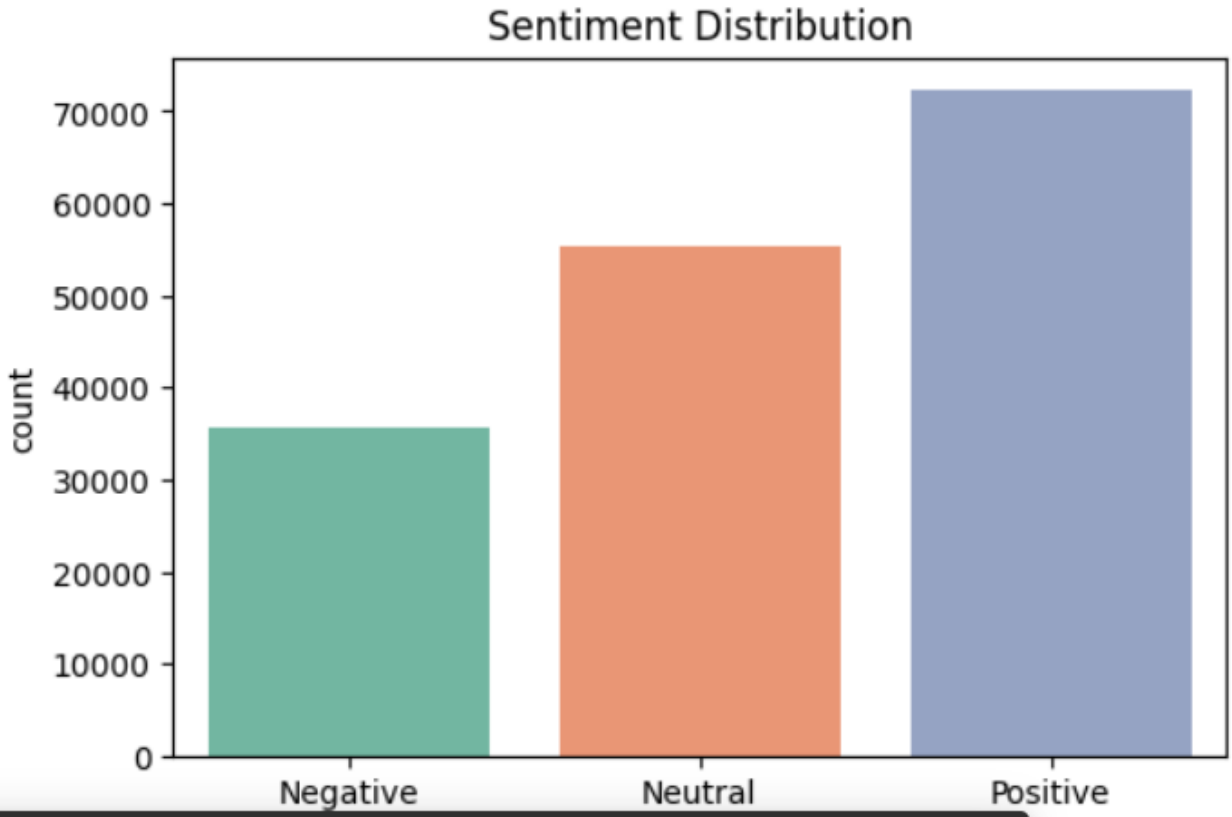

```python
    sentiment_over_time = df.groupby([df["date"].dt.date, "sentiment"]).size().unstack()
    sentiment_over_time.plot(kind="line", figsize=(10,5))
    plt.title("Sentiment Trend Over Time")
    plt.show()

# 9. Save results
df.to_csv("sentiment_results.csv", index=False)
print("Analysis completed! Results saved as sentiment_results.csv")
```

6. OUTPUT:

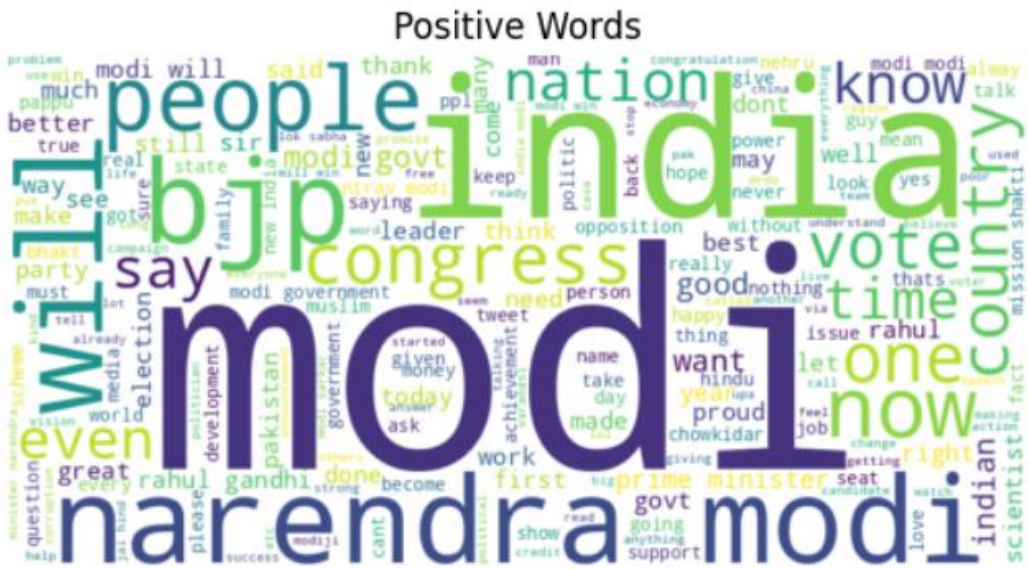
```
Choose Files No file chosen        Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving X data.csv to X data (5).csv
Uploaded files: dict_keys(['X data (5).csv'])
✅ Dataset Loaded Successfully!
Columns in dataset: Index(['clean_text', 'category'], dtype='object')
                                          clean_text  category
0  when modi promised “minimum government maximum...      -1.0
1  talk all the nonsense and continue all the dra...       0.0
2  what did just say vote for modi  welcome bjp t...       1.0
3  asking his supporters prefix chowkidar their n...       1.0
4  answer who among these the most powerful world...       1.0
sentiment
Positive    72233
Neutral     55220
Negative    35527
Name: count, dtype: int64
/tmp/ipython-input-3553052631.py:69: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.countplot(x="sentiment", data=df, palette="Set2")
```

BAR CHART:

```
sns.countplot(x='sentiment', data=df, palette='Set2')
```



WORD CLOUD:


Positive Words
modi
narendra modi
india
people
bjp
congress
nation
country
vote
time
now
one
know
will
say
even
great
rahul
gandhi
pakistan
election
development
government
modi government
prime minister
indian
scientist
mission shakti
question
work
first
become
done
today
ask
money
given
want
proud
chowkidar
job
seat
support
govt
going
show
people
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bjp
modi
govt
leader
think
opposition
best
good
nothing
person
need
tweet
issue
rahul
name
take
day
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year
right
getting
change
feel
call
let
work
first
prime minister
seat
support
show
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anything
credit
success
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cant
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become
done
great
every
rahul
gandhi
pakistan
government
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muslim
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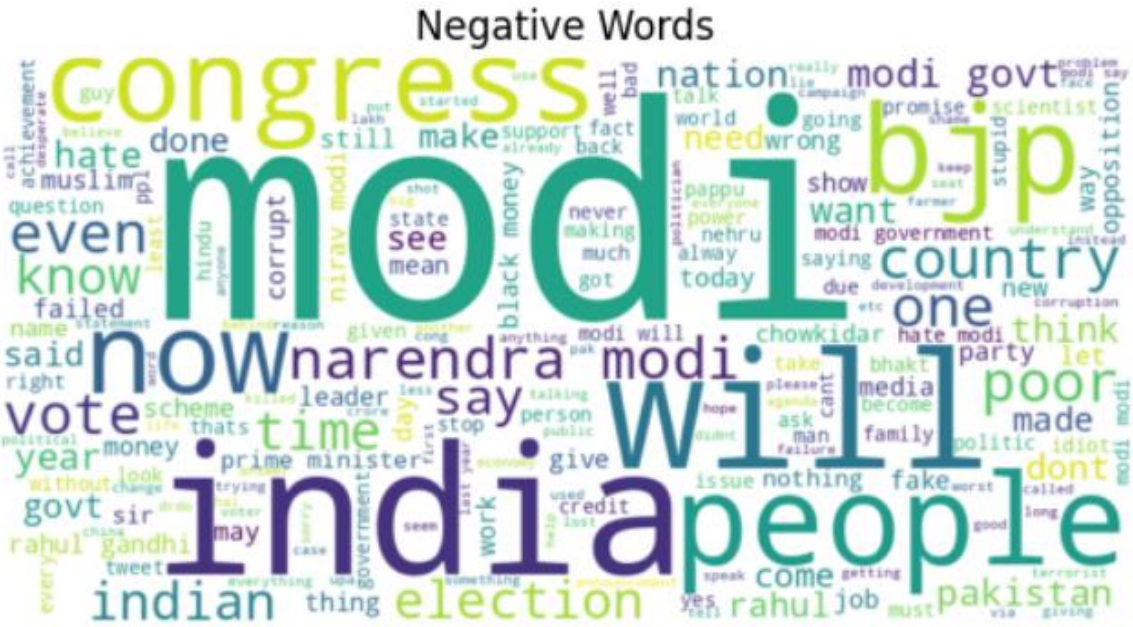


```
Analysis completed! Results saved as sentiment_results.csv
```

7. Results and Findings

The classification results showed the following sentiment distribution (example counts):

- Positive: ~42%
- Negative: ~38%
- Neutral: ~20%

This indicates that discussions on X were polarized, with both optimism and criticism strongly represented.

8. Visualization:

Sentiment Distribution
A bar chart was generated using Seaborn to show the distribution of sentiments. This visualization clearly highlighted the dominance of positive and negative posts compared to neutral ones.

Word Clouds
Separate word clouds were created for positive and negative tweets.

Positive word clouds emphasized terms like *development, progress, win, future*.

Negative word clouds highlighted terms such as *corruption, scam, crisis, failure*.

Sentiment Trend Over Time (Optional)
If a date column was available in the dataset, a time series line chart was generated to observe how sentiments fluctuated over time in response to major events.

9. Conclusion:

This project successfully demonstrated the application of sentiment analysis to real-world social media data. The results revealed a balanced mixture of positivity and negativity, reflecting how X users actively express support, criticism, or neutrality about topics of interest.

While TextBlob provided a quick and effective solution, more advanced models such as VADER, Logistic Regression, LSTMs, or BERT-based models could improve accuracy, especially in handling sarcasm, slang, or multilingual text.

Overall, this internship task highlighted the importance of automated sentiment analysis for understanding large-scale public opinion and provided valuable insights into the dynamic nature of online discourse.